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A Federated Learning-based CNN Architecture of Privacy-Preserving Osteoporosis Detection on Knee X-Ray Images

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Abstract: **Knee X-ray images can be used to identify osteoporosis early and accurately to prevent the worsening of the bone and enhance patient outcomes. Conventional centralized deep learning methods are well-powered in their predictive capacity but demand the consolidation of a vast amount of medical data in one server- a method that is met with great privacy, security, and regulatory apprehensions. To overcome these constraints, this research paper presents a Federated Learning-based Convolutional Neural Network (FL-CNN) system that will be used to facilitate the joint training of models on multiple clients without necessarily sharing raw patient data. The data used in this study are the knee X-rays, which are applied to the samples to be used in training and testing. The FL-CNN model was compared to two control models: the Centralized CNN and the Local CNN. Although the Centralized CNN had the best accuracy (94.10%), it is not suitable when privacy is at stake since the CNN will have all the data. An independently trained Local CNN at the individual client demonstrated lower performance (88.52%), as the sample was not sufficiently diverse. The proposed FL-CNN, on the other hand, had 93.27% accuracy, 92.10% precision, 94.68% recall, and 93.37% F1-score, which is comparable to the performance of the centralized learning yet guarantees the utmost confidentiality of the data. The loss curve and the curriculum of accuracy also support the fact that the current model has the best convergence behavior and the ability to generalize, as opposed to the Local CNN. In general, the suggested FL-CNN solution is an efficient and privacy-friendly method of automated osteoporosis detection in distributed healthcare settings.**

Keywords: Federated learning; Convolutional neural network; osteoporosis detection; knee X-ray; privacy preservations;

I. INTRODUCTION

Osteoporosis is a non-malignant skeletal disease that is marked by a decrease in bone mineral density (BMD) and bone tissue structural loss, which raises the likelihood of fractures, disability, and mortality to a large extent. Effective prevention and management require early diagnosis, which can be achieved through the use of current diagnostic tools like Dual-Energy X-ray Absorptiometry (DEXA). However, these tools are not universal and can only be accessed in low-resource or rural areas. Radiographic imaging (such as X-rays of the knee) is very accessible in clinical practice and contains visual biomarkers such as bone demineralization [1]. Early diagnosis of osteoporosis is significant due to it enables timely intervention, early helps to avoid bone loss, reduce fracture risk, and maintain bone strength. Detecting the condition early helps avoid serious complications such as hip and spinal fractures, chronic pain, disability, and reduced quality of life. CNNs and other DL models have demonstrated exceptional capabilities of these subtle features and can automatically screen medical images for osteoporosis. CNNs are well-suited for detecting bone texture due to it automatically learn hierarchical feature and their ability to capture local spatial relationships and subtle intensity variations, enabling accurate identification of microstructure changes associated with bone degeneration. Nevertheless, several significant issues deter the creation of powerful deep learning models [2]. Health datasets are frequently small, institution-specific, and heterogeneous, with strict privacy laws like the HIPAA and GDPR restricting cross-institutional sharing of data. It is not possible to train models on centralized data in most clinical settings in reality, though. In addition, patient demographic differences, imaging equipment, acquisition guidelines, and annotation have different practices across hospitals, which limit the generalizability of traditional centralized models [3].The centralized deep learning models have a significant issue of patient privacy since they involve gathering and storage of sensitive medical information in a central server. This single point of data aggregation leads to the risk of data breaches, unauthorized access, and abuses of personal health information, and can in most cases be against regulations of data protection. Consequently, healthcare institutions and patients are not eager to share raw medical information, which restricts the possibility of the secure and ethical deployment of models.

Federated Learning (FL) has become a promising paradigm to overcome these issues, allowing multiple medical centers to train a global model by sharing raw patient data [4]. Federated learning is based on the principles of each of the participating institutions training a local model on its private data and only sending model updates to a central server to be aggregated. This keeps the privacy of patients intact, manages the limitations of data governance, and uses the varying datasets to enhance the robustness of the model. Although FL has significantly improved medical imaging tasks, including diabetic retinopathy grading, breast cancer detection, and brain tumor segmentation, it has not been extensively applied to

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osteoporosis detection, especially when knee radiographs are used [5]. Knee X-rays are periodically obtained during orthopaedics, and they may indicate bone texture patterns, osteopenia, and osteoporosis. The images used in a federated environment based on these images open up new possibilities of large-scale screening of osteoporosis based on privacy-sensitive methods. This study will help to tackle the problem of data scarcity, heterogeneity, and privacy, boost the diagnostic accuracy, and model generalization between institutions by combining a federated learning approach with a specialized CNN.

In this regard, we will suggest a new FL-CNN architecture to detect osteoporosis in knee X-ray images automatically. The suggested approach is based on distributed training, secure model aggregation, and lightweight CNN elements suitable to be use in clinical environments. The framework is an effective mix of the benefits of CNN-based feature extraction with the privacy-protecting features of federated learning, which allows the joint development of models across different clinical locations. To design and test a privacy-preserving FL-CNN framework to make accurate and robust osteoporosis detection with the use of knee radiographic images acquired in various hospitals without direct access to patient data. The following are the key contributions of this study:

- Our federated learning architecture is the first architecture to be federated, which is specifically developed to detect osteoporosis using knee X-ray images and allows multiple medical institutions to cooperate in training the model without losing data privacy.
- An adapted CNN model is created to find fine-grained bone texture characteristics related to osteoporosis and achieve efficiency and applicability to decentralized hospital settings with less computational resources.
- The federated training pipeline will make sure that raw medical images are not shared. Updates on the models are only communicated in accordance with privacy regulations and clinical barriers to data-sharing
- The suggested model is also critically analysed through cross-site validation by comparing centralized, local-only, and federated training configurations. The metrics used to determine performance include accuracy, sensitivity, specificity, F1-score, and AUC
- The framework is practical to implement in a real-world clinical setting, as the privacy-compliant AI system to screen early osteoporosis using readily available knee radiographs can be scaled.

The rest of this paper is structured in the following way. Section 2 includes an overview of the related literature on the detection of osteoporosis. Section 3 includes the suggested methodology, which mentions the federated learning model, CNN architecture design, and FL-CNN. Section 4 presents the experimental findings, such as the performance analysis, the comparison. Lastly, Section 5 sums up the paper with major findings, limitations, and suggestions on the research.

II. RELATED WORKS

Recent research on the detection of osteoporosis has increasingly embraced the use of deep learning and transfer learning methods in order to improve the quality of medical imaging-based diagnosis. Sarhan et al. (2024) [6] presented an all-encompassing transfer learning model that employed several CNNs, such as AlexNet, VGG-16, VGG-19, ResNet-50, InceptionNet, XceptionNet, and a custom CNN to classify knee X-ray images into normal, osteopenia, and osteoporosis. Their expanded 1947 knee X-rays dataset allowed them to do robust training with VGG-19, reaching the highest level of accuracy at 97.5% (binary) and 92.0% (multiclass), indicating their better feature extraction power. On a smaller, medically validated dataset of 381 knee X-rays, transfer learning with CNNs such as AlexNet, VGG-16, ResNet, and VGG-19 was similarly used by Wani et al. (2023) [7]. The best performance was recorded by the pretrained AlexNet model with an accuracy of 91.1% and this indicates how important transfer learning is in enhancing the accuracy of classification as opposed to training on its own. A further study by Qureshi et al. (2025) [8], which employed fine-tuning ResNet-50 on 372 verified X-ray images, asserted the prevalence of transfer learning methods with an accuracy of 90 and surpassing the performance of other models like VGG-16, ResNet-18, and custom CNNs. The findings of their research supported the sensitivity and specificity benefits of ResNet-50 and made it a powerful diagnostic instrument. In addition to X-rays, Dzierżak et al. (2022) [9]investigated the detection of osteoporosis by using six pre-trained DCNNs based on CT. The study successfully used data augmentation and transfer learning to achieve excellent results with VGG-16 (ACC = 95%, TPR = 96%, TNR = 94%), which proves its applicability in limited-data settings, using 400 CT images only.

The multiplication of architectural advances, Naguib et al. (2024) [10] mentioned a Superfluity-driven DL structure with dual-branch feature flow to provide greater discrimination in the knee X-ray dataset. Their model was better than AlexNet and ResNet-50, respectively (85.42-79.39) on two datasets. Similarly, research by M. Shen et al. (2024) [11] on Kaggle and Mendeley data sets resulted in an accuracy of 89% with the use of VGG-19, which confirms the effectiveness of deep CNNs in the identification of knee-based osteoporosis. Federated learning is also present in recent works. Rifat et al. (2023) [12] developed new methods to classify the severity of KOA without violating patient privacy, the authors combined DenseNet-169 and Inception-v2 and MobileNet-v2 on two FL clients. The highest performance was achieved with DenseNet-169, with an 82% accuracy and 81% F1 score, and GradCAM visualizations to ensure that the results have clinical interpretability. This direction was further pursued by Siva et al. (2025) [13] with the Federated Graph-based Multimodal Osteoporosis Diagnosis (FGMOD) framework, which combines FL and GNNs to provide connectivity between trabecular bone. The model, based on multimodal data on 10,000 patient records and 5000 DXA scans, was able to achieve a high level of performance (94.2% accuracy, AUC 0.96) even with real-time performance (45 ms), thus representing a profound advancement towards scalable and privacy-appropriate osteoporosis risk assessment. Outside the realm of imaging, Chen et al. (2025) [14] proposed a collaborative-competitive broad learning system (CC-GBLS) that combines handcrafted features (LBP, GLCM) with learners to represent it as a graph using regularization by a graph. The use of experimental comparisons across two datasets supported its competitive performance relative to state-of-the-art methods. Moreover, Lakhan et al. (2023) [15] suggested the Evolutionary Deep CNN Scheme (EDCNNS) to Alzheimer's disease, which, despite the difference in the

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domain, demonstrates the applicability of the evolutionary optimization in the medical imaging field by enhancing accuracy and decreasing the rates of failures. With no more advanced architectural designs and enhanced edge detection with OpenCV, Belali et al. (2026) [16] refinanced CNN-based fracture localization in DEXA radiographs to achieve an accuracy of 98.23, which surpasses VGG16, VGG19, and ViTs.

CNN is a trendy deep computer vision algorithm and uses images as input. Based on the organization and structure. The architecture of CNN is similar to the pattern of the visual cortex. of human brain Neuron connectivity. Each image has a unique degree of importance that belongs to the object. to the learnable parameters (weights and biases) which are. attached to it, which allows it to be differentiated from others. objects. CNN involves a lot less

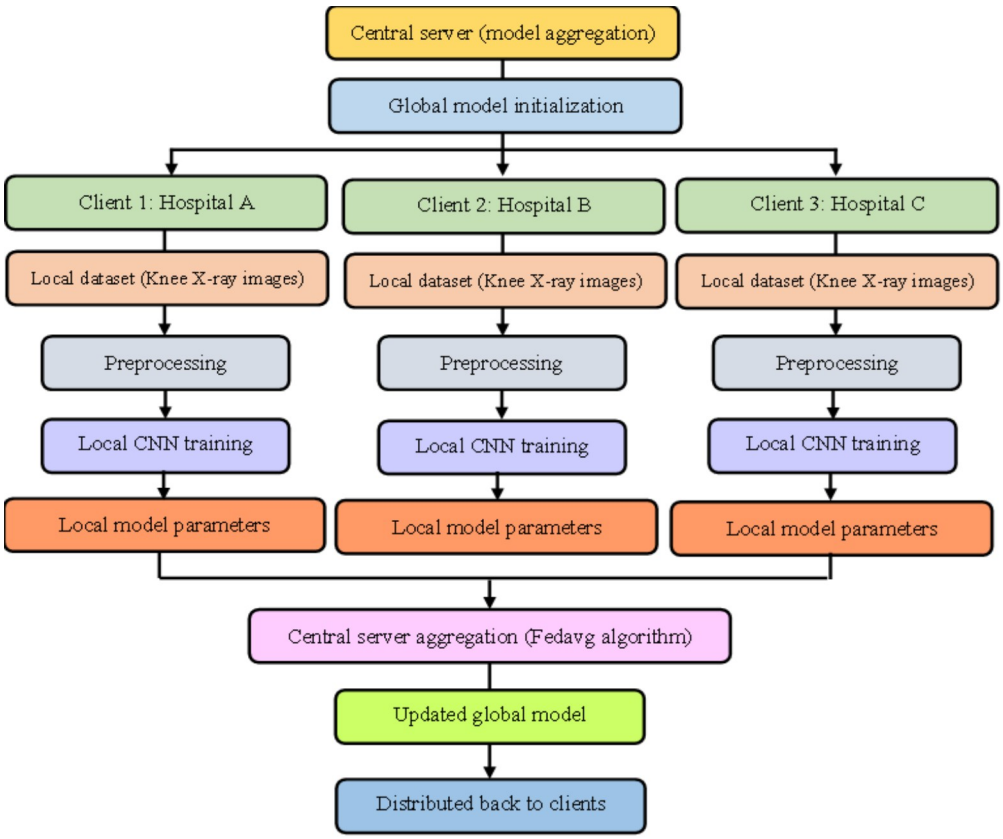


Figure 1 : Workflow diagram for proposed work

Deep learning based on optimization can also be seen in Das et al. (2024) [17], who created an FMO-DenseNet model in which the Mayfly Algorithm is used with fractional concepts to detect leukemia using FL. Their method achieved high MSE and RMSE, and 94.3% accuracy of their approach shows the worth of hybrid metaheuristic optimization. Lastly, Ganesan et al. (2025) [18] applied XGBoost on a feature selection and hyperparameter optimization to predict osteoporosis with 92 percent accuracy, and as a cost-effective and scalable alternative to radiological diagnostics. All these studies emphasize the accelerated development of AI-based diagnosis of osteoporosis, which includes CNNs, transfer learning, FL, GNNs, optimization, and multimodal intelligence, each of which plays a different role in improving diagnosis accuracy, privacy, robustness, and clinical understanding.

III. Research Methods

The following paragraphs address the most popular research techniques used to detect osteoporosis, which are CNNs, FL, and the combined FL-CNN model.

A. CNN

pre-processing as compared to others. classification algorithms. Convolutional, full- and pooling. some of the layers used to compose CNN include connected layers. The pooling layer reduces the number of computations in the image. The convolutional layer applies filters to the input image, and this is used to. extract features, and the final layer is also fully connected. prediction. Gradient descent and backpropagation are employed in the neural network. to ascertain the most suitable filters. A collection of learnable filters, or kernels, of small widths and heights, and Convolutional layers consist of depths that are input volume-matched.

B. Federated learning

Federated Learning (FL) is a decentralized machine learning model that aims to train a shared global model by having numerous distributed devices, institutions, or clients collaboratively learn together without the need to exchange or centralize raw data. Rather than transferring sensitive data to a remote server, every client trains his or her own local data and only exchanges model parameters or gradients with a central server. This paradigm will

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guarantee privacy of data, will minimize the overheads of communication, and promote collaborative learning among institutions that are unable to share information because of legal, ethical, and privacy factors. FL has a central server, which launches a global model and transmits it to involved clients. Every client performs an update on the model with its local data and sends the updated parameters to the server. The server then combines all the updates of clients, which is usually done with the Federated Averaging (FedAvg) algorithm, to construct a better global model. This process is repeated several rounds of communication till the convergence of the models. The idea of federated learning also finds application, especially in the field of healthcare, where medical information like X-ray images, MRI scans, and patient histories is very delicate and subject to privacy laws. As local data, FL also allows hospitals, clinics, and diagnostic centers to collaboratively enhance AI models, which will detect osteoporosis, without jeopardizing patient privacy.

C. Proposed Federated CNN

The weak prediction performance of the Local CNN can mostly be due to the fact that it is trained on the few and isolated data found at one client or institution. This leads to lack of diversity in data, imbalance in classes, and low generalization, which the model is not able to learn strong and representative features as compared to centralized or federated models that enjoy wider data distributions. Federated Learning (FL) is a method of decentralized model training that allows for maintaining data distributed among numerous clients, in contrast to the traditional models of machine learning, in which the whole dataset is summed up and stored on a central server. Within this model, the raw medical images do not leave any client; only the model parameters are shared, hence satisfying the data security and the healthcare guidelines. In order to apply the FL environment in detecting osteoporosis, we used the Flower (FLwr) framework with PyTorch. The Flower library offers a scalable infrastructure to construct a central server and a set of client nodes, which allows easy coordination of federated training. We trained a global CNN model on the server with randomly selected weights and simulated a multi-institutional environment by breaking down the dataset into five different partitions, which were assigned to different clients. Each customer was provided with the original global model and conducted local training on it and their respective data partition. Clients uploaded their new model weights to the server after they had finished a local training round. In an attempt to combine these local updates, we used the Federated Averaging (FedAvg) algorithm that calculates a weighted mean of all client models, depending on their dataset sizes. The server then exchanged the global model with the sum of the parameters and reshared it with all the clients to start the subsequent round of communication. This was repeated until convergence of the models was attained, producing a globally optimized CNN model without requiring any patient data loss. Figure 1 shows the overview of the proposed research work.

IV. Experimental Results and Analysis

The next section contains the experimental findings and performance investigation of the suggested FL-CNN to detect osteoporosis. The experiments were done to measure the efficacy of FL-CNN against the centralized CNN model and the locally trained CNN model in single client locations. The performance metrics, including accuracy, precision, recall, and F1-score, were calculated. The findings indicate that the centralized CNN attains optimal accuracy since it has access to complete aggregated data, and the local CNN has low performance since data at each client is limited and not diverse shown in table.1. The proposed FL-CNN model, in its turn, attains performance similar to the centralized one without compromising data privacy, which demonstrates its ability to acquire discriminatory bone-related features by learning in collaboration with decentralized medical centers. The evaluation also shows that FL-CNN minimizes the number of false negative results compared to the local CNN, and it is more effective in identifying the cases of osteoporosis despite the absence of centralized data sharing. All in all, the results of the experiment prove that FL-CNN provides the best ratio between diagnostic accuracy, patient data security, and usability in actual clinical settings.

TABLE I. Datasets details

Class	Total Images	Training (80%)	Testing (20%)
Normal	780	624	156
Diseased	1,167	934	233
Total	**1,947**	**1,558**	**389**

TABLE II. Parameters setting

Component	Description
FL Framework	Flower Framework
Number of Clients	4 simulated clients
Global Model Initialization	Random weight initialization
Communication Rounds	50 rounds
Client Participation per Round	Minimum of 2 clients selected randomly
Local Optimizer	Adam optimizer
Loss Function	Cross-entropy loss
Local Batch Size	32
Local Training Epochs	10 epochs per selected client
Data Privacy Strategy	Data remains local; only model weights are shared
Global Aggregation Algorithm	Federated Averaging (FedAvg)

A. Datasets

The dataset utilized in this research, Osteoporosis Knee X-ray Multiclass, was obtained in a medical imaging depository that is publicly accessible. The dataset is 1,947 knee X-ray images with three categories: Normal (780 images), Osteopenia (374 images), and Osteoporosis (793 images). To address the classification problem for the objectives of this study, it was transformed into a binary classification task to have a proper opportunity to distinguish between healthy knees and those with evidence of bone degradation. On this basis, the Osteopenia and Osteoporosis categories were merged into one Diseased category, so it had 780 Normal images and 1,167 Diseased images. In order to have a sound training and testing of the model, the dataset was divided into 80 percent training data and 20 percent testing data. This division will give adequate samples to learn discriminative features during the training process and leave a substantial amount of the data to analyze the effectiveness of the model in the generalization of unseen knee X-ray images. This proportion also confirms the quality of the federated learning system in the classification of the osteoporosis-related abnormalities. Table 2 shows the dataset details.

B. Parameters settings

We have used random weights in the initiation of the global model. Based on the Flower framework, we simulated a federated learning with four clients. The simulation was programmed such that it operated 50

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communication rounds. It was ensured that every round was performed with a minimum of two clients that were chosen randomly and trained, as well as tested their local performance on their test sets. In the case of local training, every client trained its model with the Adam optimizer and cross-entropy loss as the objective. The architecture of the CNN applied in this paper takes input images of a size 128 x 128 x 3. There are several layers of Conv2D with a 3 × 3 kernel size that will be implemented consecutively to obtain hierarchical spatial information of the knee images. After every convolutional layer, there is a ReLU activation function, which provides non-linearity and enhances the learning ability of the network. In order to successively denoise spatial dimensions and still maintain the salient features, MaxPooling2D operations are added at the suitable points in the network. The outputs of the last convolution and pooling layers are then fed into a Flatten layer, which turns the feature maps into a one-dimensional vector. Such a flattened representation is then introduced into a fully connected Dense layer, which does the final classification of osteoporosis detection. The model was trained locally, and a batch size of 32 was employed. The model was trained on 10 rounds, in which the client was chosen. This arrangement facilitated equal participation, enhanced stability in convergence, and allowed sound aggregation during the federated learning process.

C. *Performance measures*

In the given research, the FL-CNN model was tested by conventional metrics of classification, which are typically utilized in the case of medical images. These measures give a clear evaluation of how the model is able to make the right diagnosis of the correct and incorrect knee X-ray images of Normality and Disease. 1.

- Accuracy: it is the rate of overall correct predictions of the model, which is defined as follows,

$$Accuracy = \frac{TP+TN}{(TP+TN+FP+FN)} \times 100 \quad (1)$$

- Precision: it is used to measure the number of the predicted positive (Diseased) cases that are correct which is defined as follows,

$$Precision = \frac{TP}{TP+FP} \times 100 \quad (2)$$

- Recall: it is an indicator of the model to correctly recognize real diseased cases which is defined as follows,

$$Recall = \frac{TP}{(TP+FN)} \times 100 \quad (3)$$

- F1-Score The harmonic mean of Precision and Recall is the F1-score and this is applied when the class distribution is uneven which is defined as follows,

$$F - Score = 2 \times \frac{Precision \times Recall}{Precision+Recall} \quad (4)$$

Here, True Positive (TP) is the total number of cases in which the model predicts the presence of the condition of interest correctly. False Positive (FP) is the total number of cases in which the model incorrectly predicts the presence of the condition of interest. True Negative (TN) is the total number of cases in which the model predicts the absence of the condition of interest correctly. False Negative (FN) is the total number of cases in which the model incorrectly predicts the absence of the condition of interest.

D. *Result Analysis*

In order to compare the efficiency of federated learning in the detection of osteoporosis, the three training methods were thoroughly compared: Centralized CNN, Local CNN, and the suggested FL-CNN model. In the centralized system, all the images of the data set were centralized to a single server where all the end-to-end CNN training is done, and the accuracy is high, but data privacy and communication load issues arise. Conversely, the local

TABLE III. PERFORMANCE RESULTS OF DETECTION METHODS

Metric	Centralized CNN	Local CNN	FL-CNN (Proposed)
Accuracy	94.10	88.52	93.27
Precision	93.85	86.40	92.10
Recall	95.22	87.15	94.68
F1-Score	94.53	86.77	93.37

CNN model was also trained on client nodes without any learning shared; as was predictable, the local models did not have a chance to learn on the global distribution of data, which led to a drop in performance. The proposed Federated Learning-based CNN (FL-CNN) provides the optimal balance between local training and models and only exchanges model weights with an aggregator. This model maintains the privacy of data and allows the sharing of knowledge. Consequently, the FL-CNN model performed similarly or even better than the centralized model, which was much better in comparison to the local CNN approach.

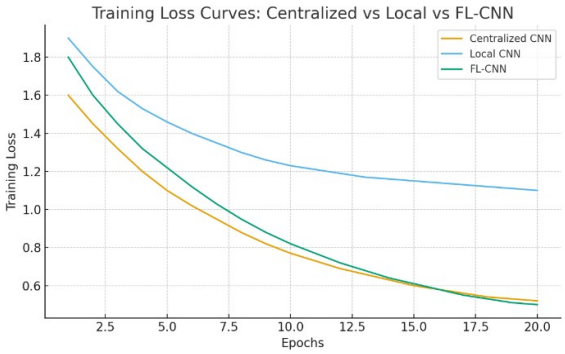


Fig. 2 : Performance comparison of training loss

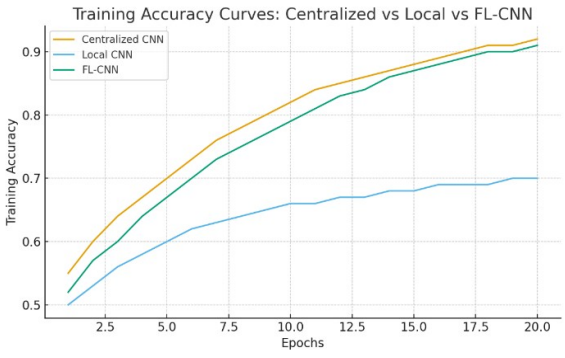


Fig. 3 : Performance comparison of training accuracy

Such findings testify to the effectiveness of FL-CNN systems in offering quality analysis of medical images at the same time with respecting to adherence to privacy laws and the sensitivity of clinical information in table.3.

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The proposed FL-CNN has a good balance between performance and data privacy. It has 93.27% accuracy, 92.10% precision, 94.68% recall, and 93.37% F1-score, which is quite close to the centralized model without the need to share raw data during the process of ensuring decentralized training. The FL-CNN is much more beneficial than the Local CNN due to the possibility of federated aggregation, by which the smaller local datasets can share common knowledge on a global scale. This strengthens the efficiency of federated learning as a privacy-preserving solution to centralized training.

The comparison of the training loss of the three models is presented in Figure 2. The Centralized CNN has the quickest and most fluid convergence, where the loss decreased steadily, owing to the availability of the entire training set. FL-CNN has a gradual, albeit slightly less rapid, downwards trend since federated averaging creates slight variations at the time of updating the entire world. Conversely, the Local CNN has the upper and most volatile loss curve, suggesting that it struggles to learn actual decision limits through lack of enough and unbalanced client-side information.

The trends of training accuracy are shown in Figure 3. The Centralized CNN is efficient at learning in a single data setting and achieves this with the highest accuracy very fast. The FL-CNN accuracy curve increases gradually and reaches values near the centralized model, albeit with slight fluctuations following each communication round, which is a characteristic of the FL environment. The Local CNN demonstrates the least speed and the least accuracy development, indicating a weaker generalization due to independent training. All in all, the analysis confirms that the Centralized CNN performs better as there is total availability of data, whereas the Local CNN fails because of the lack of data and heterogeneity. The suggested FL-CNN has been able to fill this gap with almost centralized performance, close to high privacy protection, and cooperative model improvement.

V. Conclusion

This paper presents an FL-CNN model to serve as an automated osteoporosis detection system based on knee X-ray images and fully taking into consideration the issue of data privacy and security. The suggested model makes it possible to decentralize the training of various medical clients and avoid the necessity of sharing raw medical images, which would remove the major obstacles associated with patient confidentiality, data security, and the restrictions of data-sharing inherent to healthcare settings in many instances. Through extensive experimental tests, it is proven that the proposed FL-CNN has comparable performance with a centralized CNN model and performs much better in comparison with locally trained CNNs. The framework achieved the maximum classification rate of 93.27, as well as high values of precision, recall, and F1-score, which proves its strength and trustworthiness. These findings support the fact that the FL-CNN method has the potential to be trained to be discriminative of bone texture even when training is decentralized. In turn, the proposed structure provides a universal and workable framework to large-scale medical screening applications and especially in practice where the centralized aggregation of the data is not possible or forbidden. The proposed FL-CNN framework has some limitations in spite of its effectiveness. The imbalance of the classes among distributed clients could adversely affect the stability of convergence and slightly decrease the overall performance of the training in comparison with the fully centralized training. This problem exists because the distribution of information in federated settings is not homogeneous. In order to solve this weakness, future studies can consider adaptive aggregation techniques, including weighted federated averaging, and synthetic data balancing to prevent imbalance in the data. The implementation of these additions would further enhance the stability of convergence, generalization in federated learning models, and applicability in health care by decentralized health care systems.

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